

ANNA, HANA, SREEYUTHA

Forecasting the Progression of Humanity on the Kardashev Scale through 2100 A.D with ML



A high-resolution image of Earth from space, showing the continent of South America in the center. The landmass is green with brown mountain ranges, surrounded by deep blue oceans. White clouds are visible over the oceans and parts of the continent. The Earth's horizon is visible on the left and right sides. The background is a dark, star-filled space.

Part 0: Background

Kardashev Scale

Hypothetical measure of an alien civilization's level of technological advancement

- *Kardashev wanted to classify alien civilizations based on the energy consumption they could use in interstellar communication*

$$K = (\log P - 6)/10 \quad \text{where } P = \text{energy consumption rate in Watt}$$

Type K = 0 civilization: “pre-planetary”

Type K = 1 civilization: “planetary” (planetary-scale energy control)

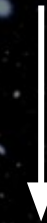
Type K = 2 civilization: “stellar” (star-scale energy control)

Type K = 3 civilization: “galactic” (galaxy-scale energy control)

- Currently, human civilization sits at $K \sim 0.72$ (BP, 2025)
- By 2100, where will humanity sit on the Kardashev scale?

Paper Methodology

For 42 countries, collect input data (GDP, population, ecological footprint....) and output energy consumption data $\equiv E_{42}$ (1960-2020)



Train Random Forest model to predict E_{42} for future 2020-2060 using projections of input features



Model historical ratio and use ARIMA to forecast $r(2020-2060) = E_{42}/E_{\text{global}}$



Solve for E_{global} (2020-2060) and compute $K(2020-2060)$

*** Methodology Clarification: RF does not directly compute E_{global} due to lack of projection data for all countries. Only include countries with major role in global economy & development)

A high-resolution image of Earth from space, showing the continent of South America and the surrounding oceans. The Earth is curved, with the horizon visible on the right. The background is a deep black space filled with numerous small, white stars. The text "Part I: Reproducing Results" is overlaid in a large, white, sans-serif font across the lower half of the image.

Part I: Reproducing Results

Random Forest Model - Predict Energy Consumption per Country

Inputs: Ecological Footprint, GDP, Population, Biocapacity, Year, Urban Population, Country Code, Temperature, Precipitation

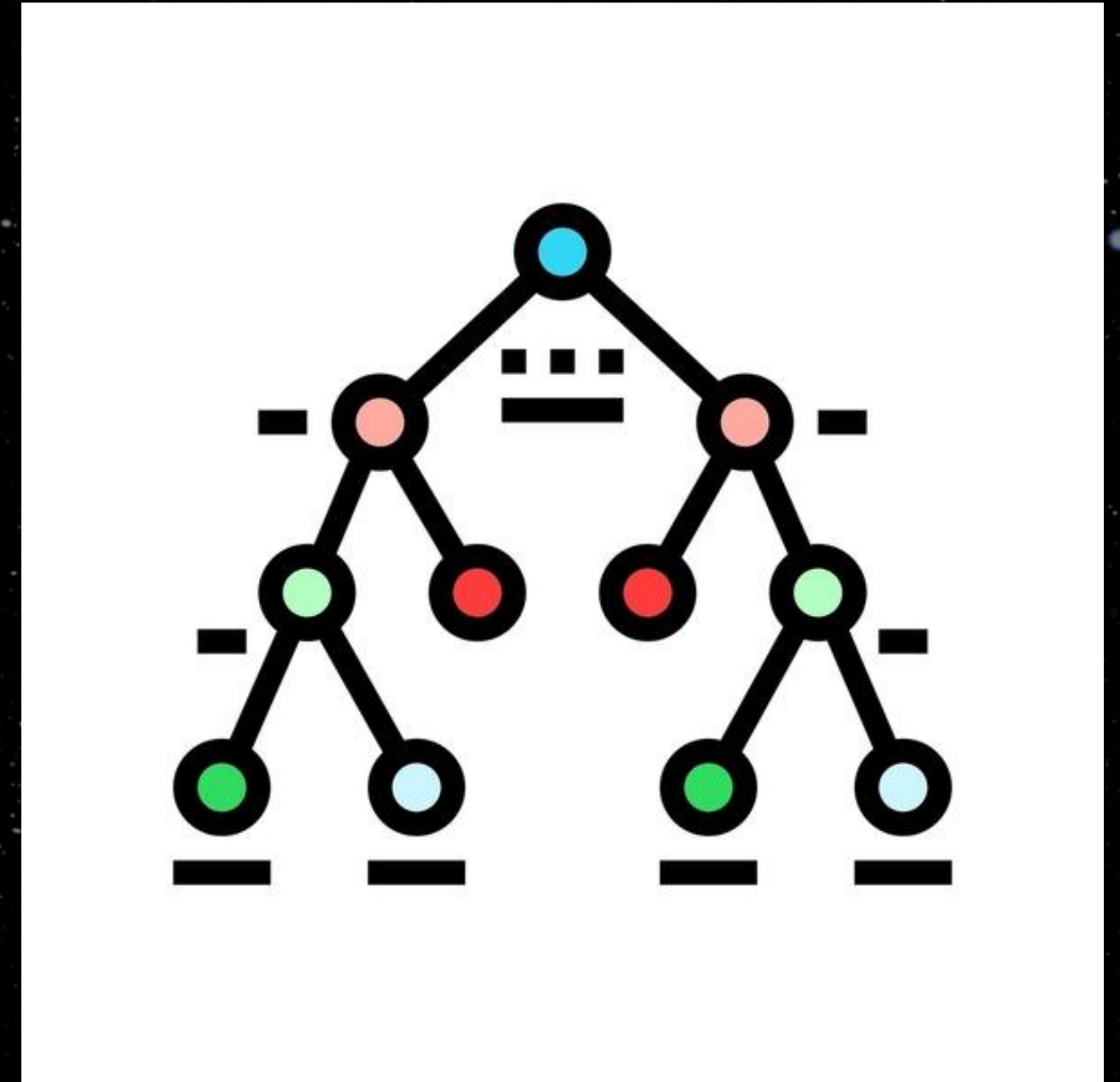
Output: Energy Consumption

Why Random Forest Model (LightGBM)?

- Captures non-linearity between input features and EC
 - trees build sequentially to fix mistakes of previous tree
- Handles high dimensionality data
- Relatively small dataset less likely to overfit
 - transformer and LSTM require more data
- Good interpretability over deep learning

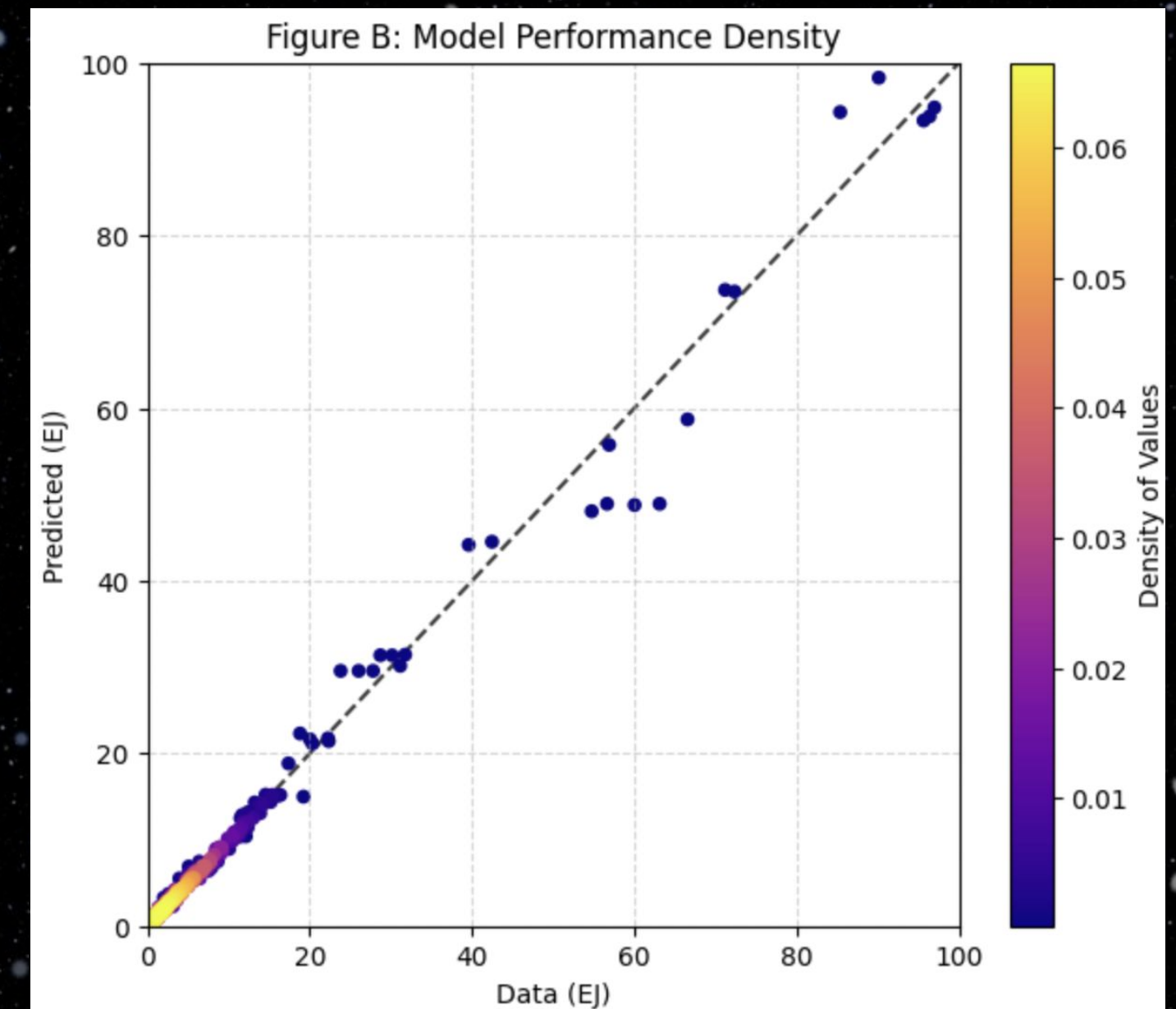
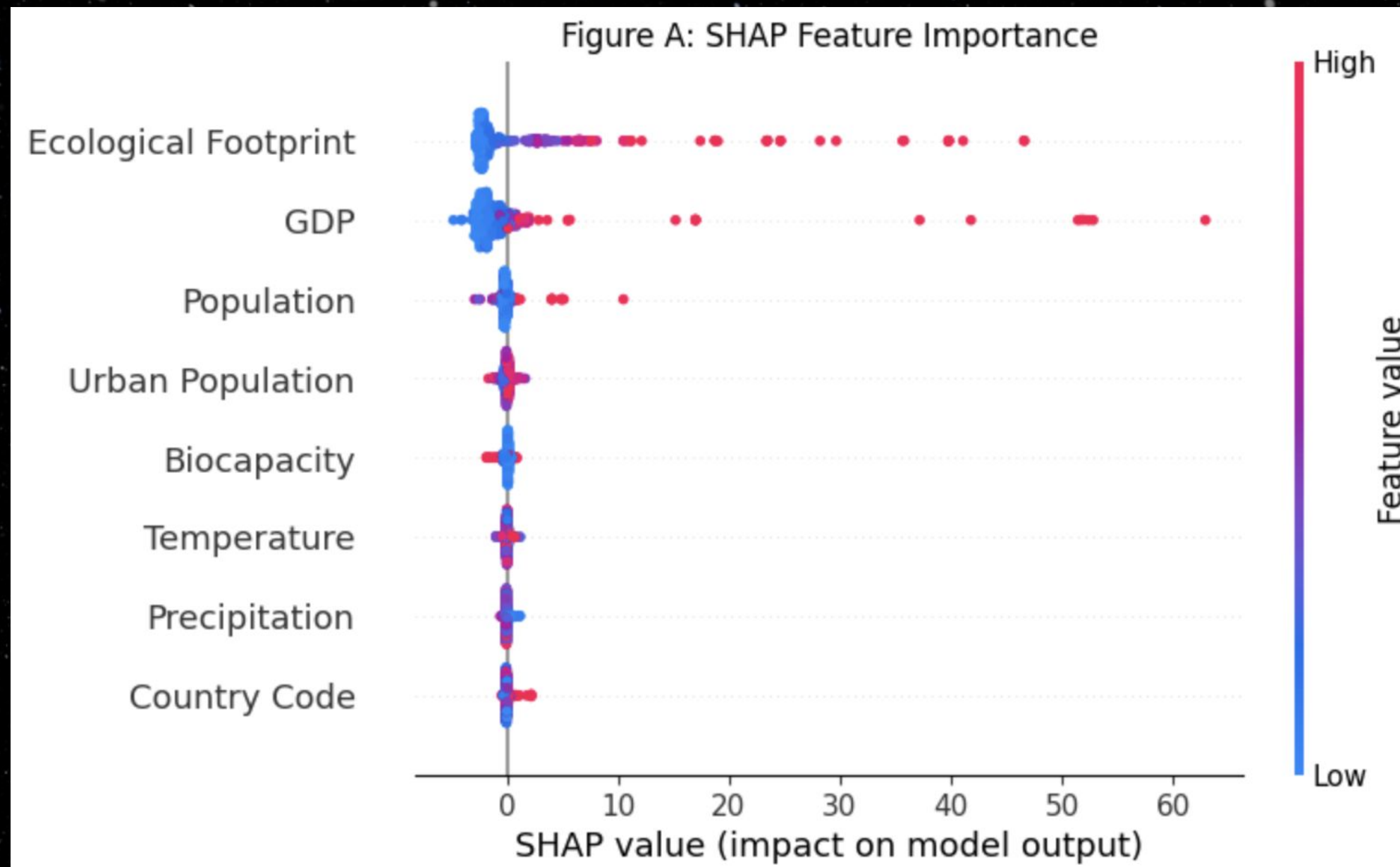
Why ARIMA?

- Good for time-series modeling



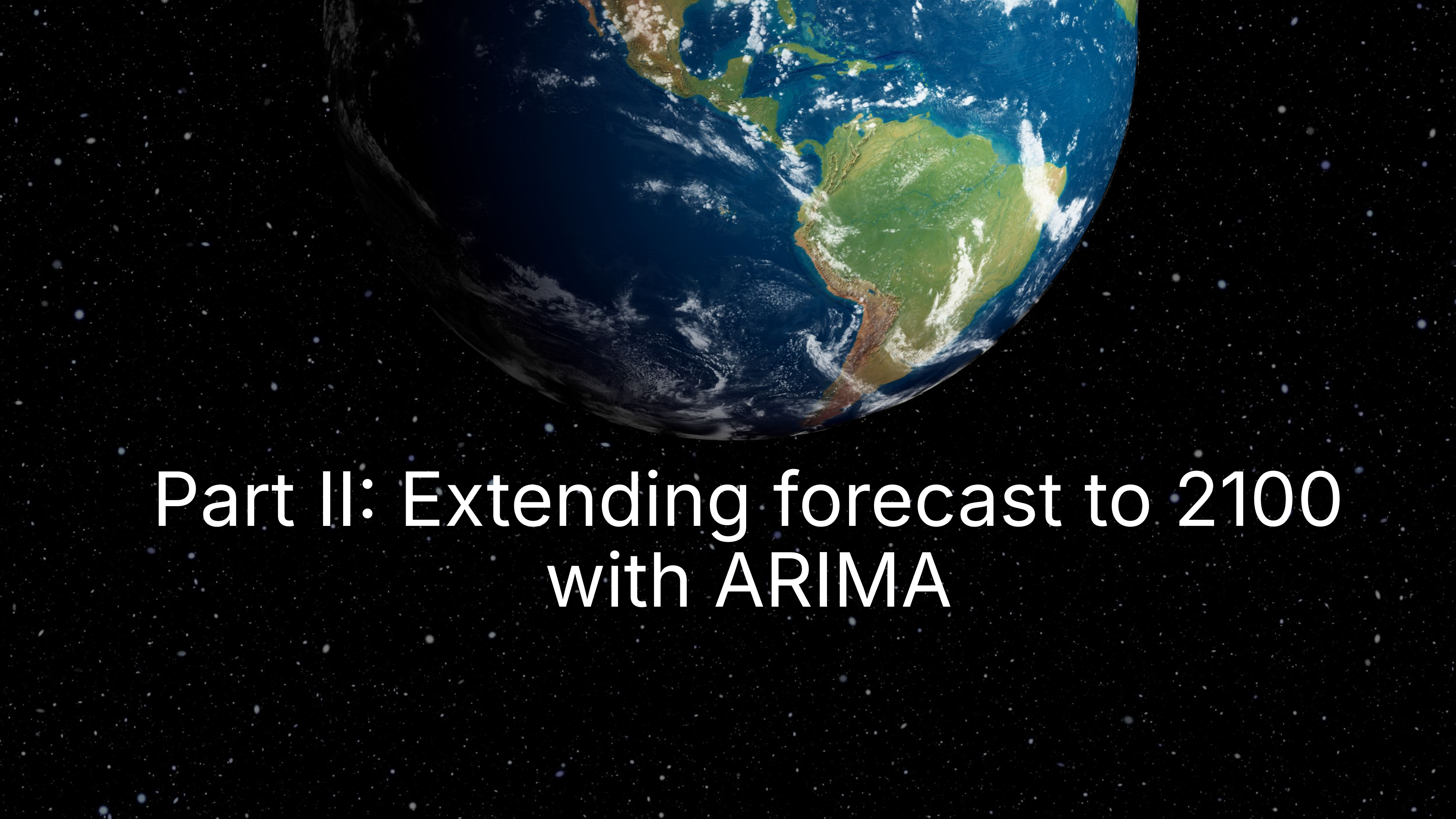
Random Forest model

- Good performance: $R^2 = 0.98$ (validation data)



Answers what drives EC the most?

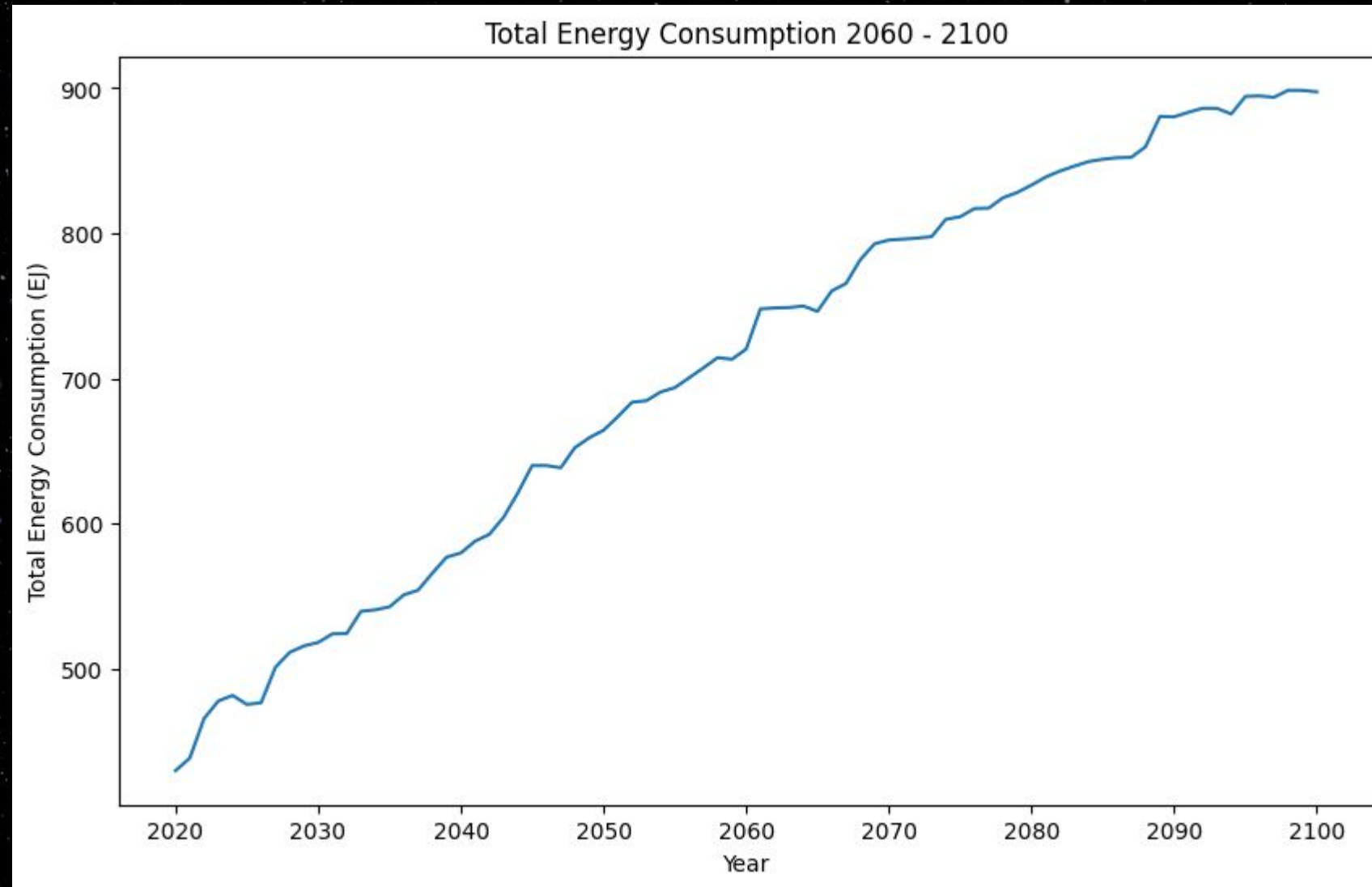
Accurate Model



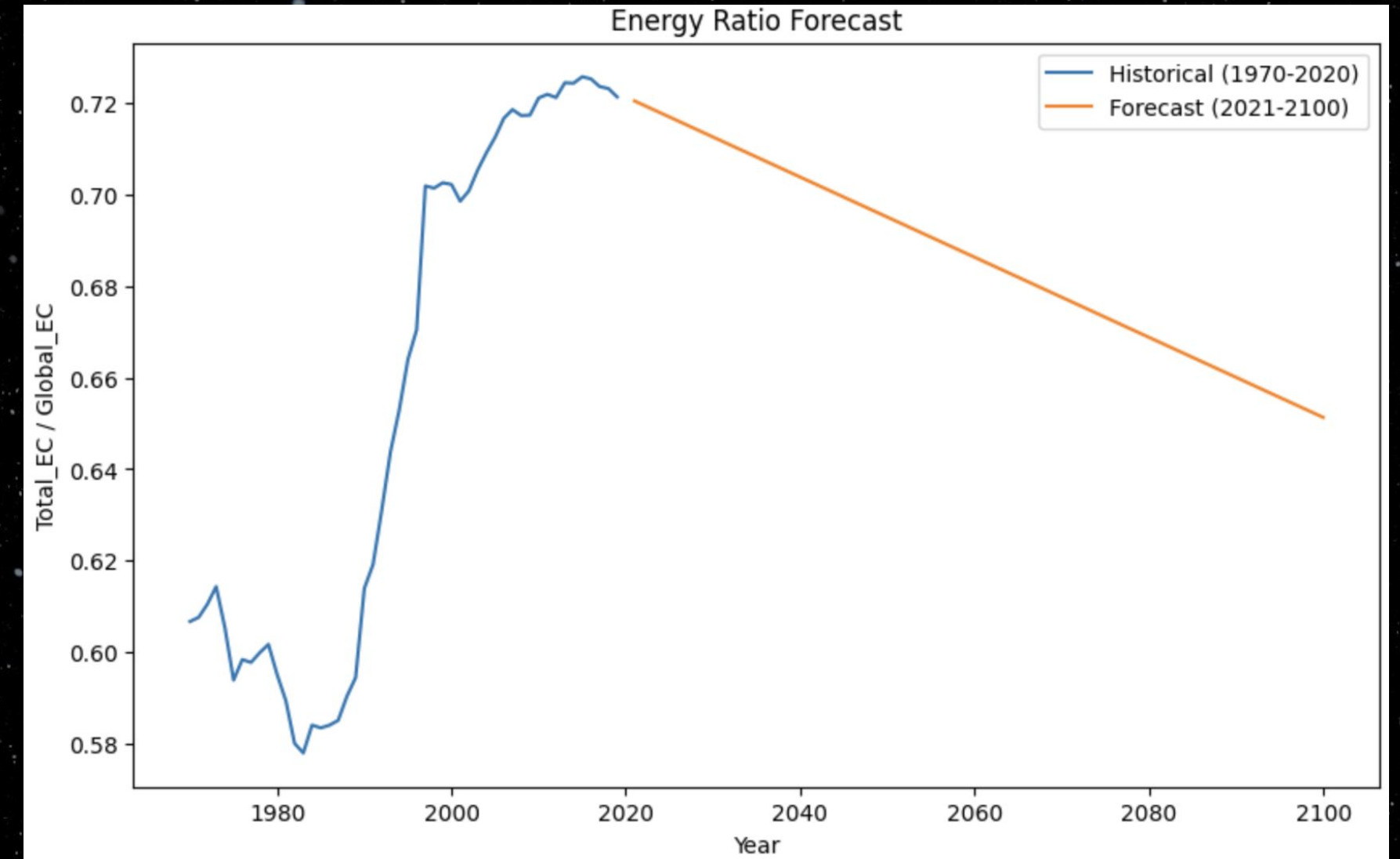
Part II: Extending forecast to 2100 with ARIMA

Kardashev Prediction for 2100

RF: E_{42} [2060-2100]

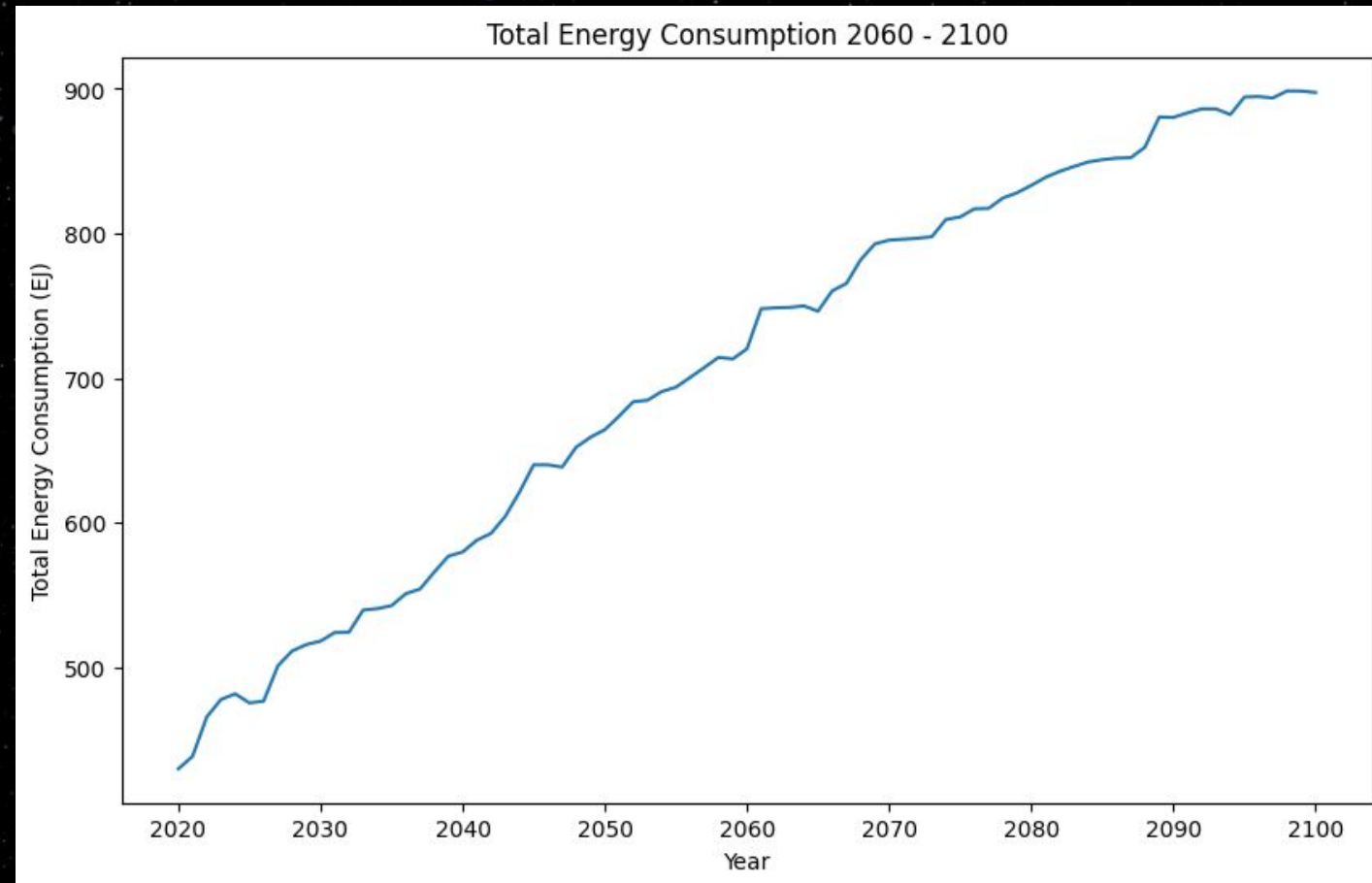


ARIMA: $E_{42} / E_{\text{global}}$ [2060-2100]

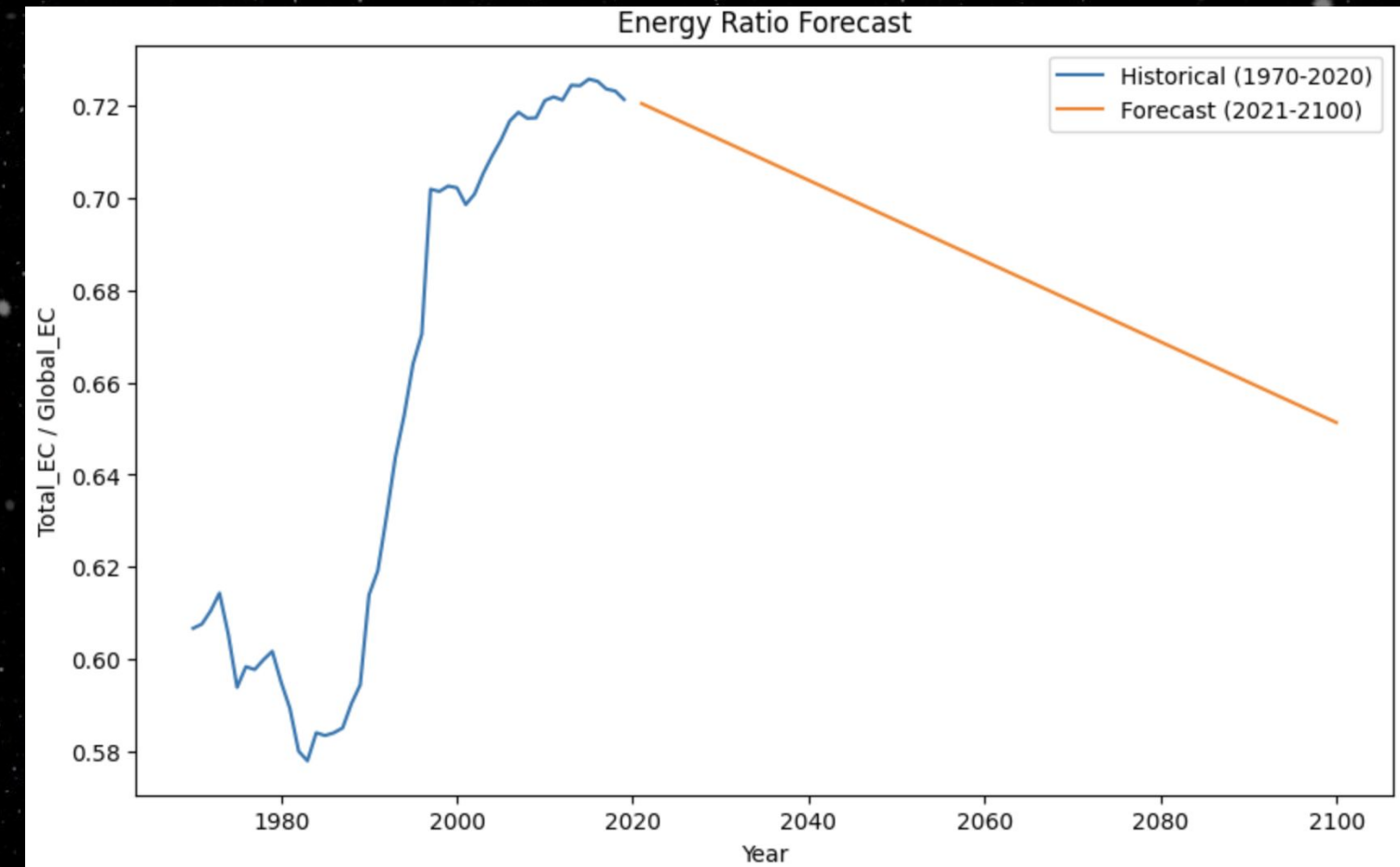


- Problem: 2060 - 2100 feature input dataset is unavailable
- Solution: trained RF on population, GDP
- Problem: ARIMA has poor performance
- Solution: Do not forecast well with a small amount of data, works better for short-range predictions, poor at predicting turning points, fails to accurately predict sudden changes

RF: E_{42} [2060-2100]



ARIMA: $E_{42} / E_{\text{global}}$ [2060-2100]



$$E_{\text{global}} [2100] = E_{42} [2100] / (E_{42} / E_{\text{global}} [2100])$$

$$\Rightarrow K [2100] \sim 0.768$$

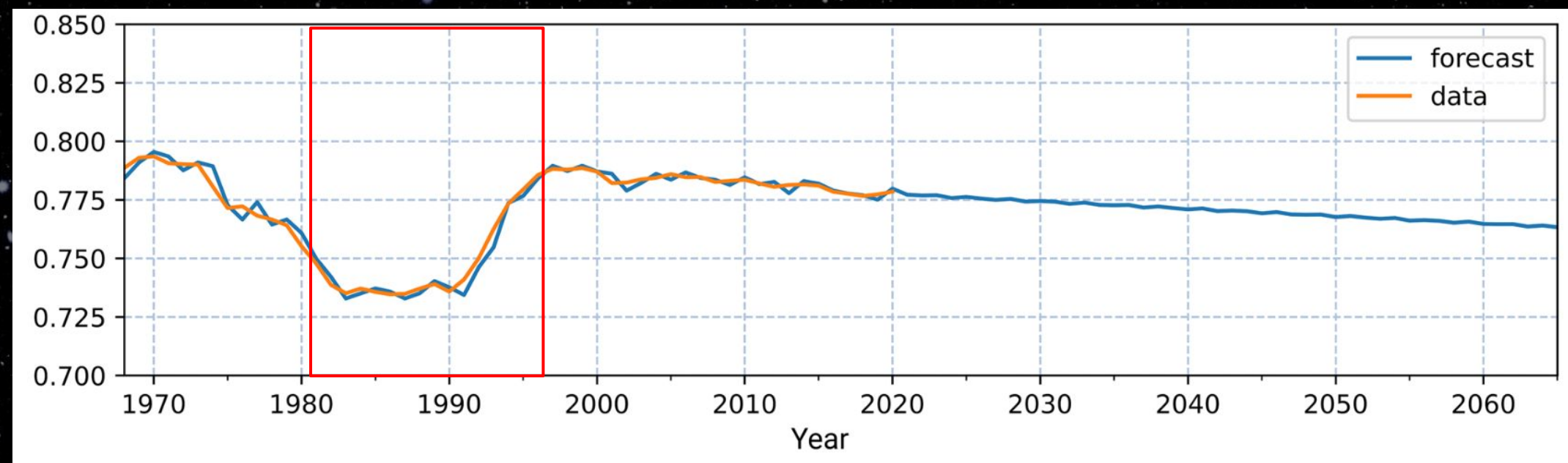
A high-resolution image of Earth from space, showing the continent of South America in the center-right. The land is depicted in shades of green and brown, indicating vegetation and terrain. The surrounding oceans are a deep blue, with white clouds swirling around the globe. The background is a dark, star-filled space. Overlaid on the lower half of the image is the text "Part III: Improving the model" in a large, white, sans-serif font.

Part III: Improving the model

Issues with ARIMA forecasting

- Predicting based on past time series, not real data (geopolitics, economics, population) → ratio is affected by real events
 - Ratio depends heavily on countries outside data set
- Largest variance will never be captured (~0.04 uncertainty)
- Not able to predict any finer features with limited data

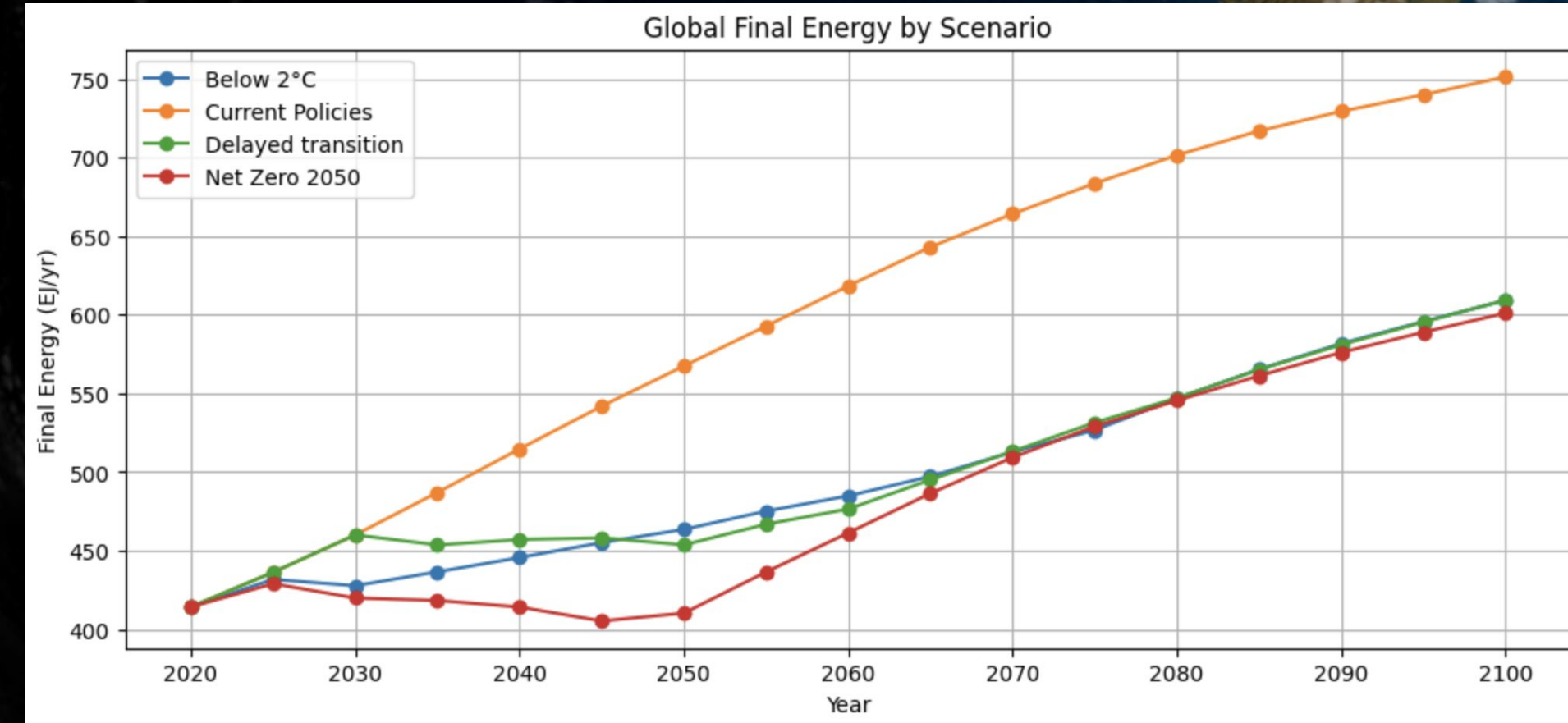
1970-80s oil shock?



Literature review

How can we incorporate real world dynamics in our model?

- Several climate models predict energy consumption to reduce drastically depending on different climate mitigation policies
- In reality, 1970s - 1990s saw large efficiency gains (energy consumption/ unit of GDP) while large absolute reductions not observed



"Energy savings rates across all sectors and in almost all countries have slowed since the late 1980s, as has the decline in CO2 emissions relative to GDP" (International Energy Agency)

→ factors that impact E_{42}/E_{global} : relative population, relative GDP, energy efficiency (climate impacts)

- Cannot predict shocks due to geopolitical conflict, must include some uncertainty from this

Forecasting energy ratio with projected data

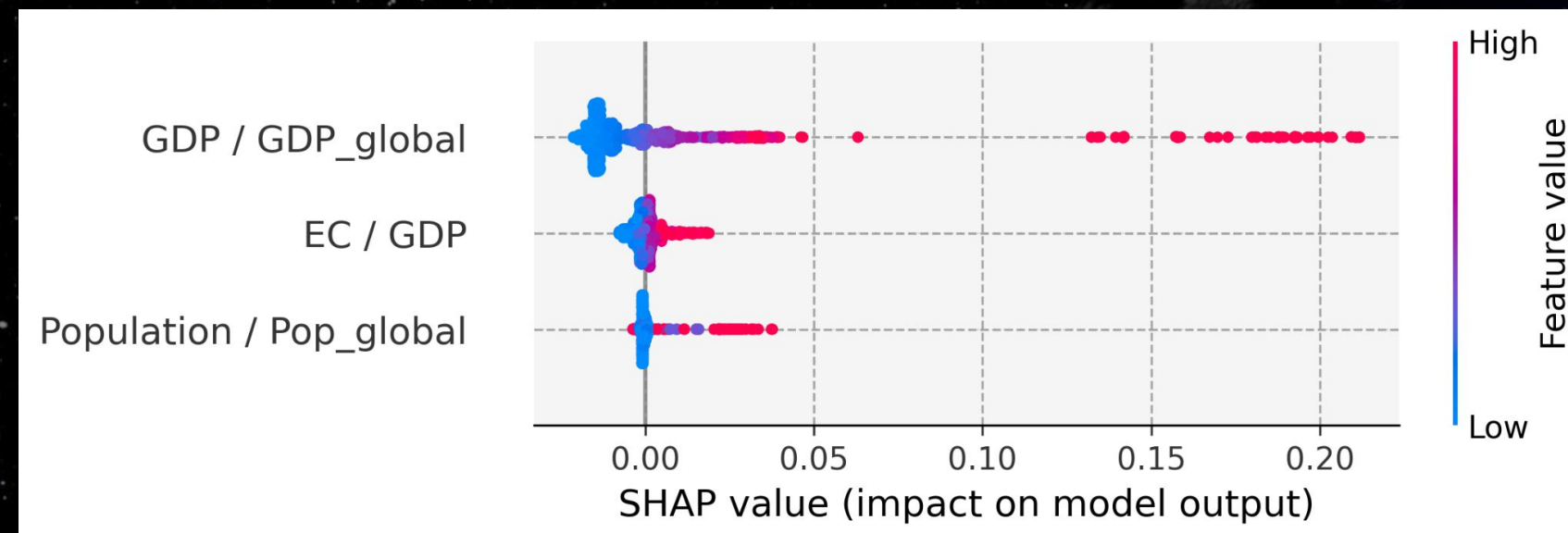
- **Input variables (1960-2025)**

- Population Share: $\text{population}_{\text{country}} / \text{population}_{\text{global}}$
- Economic Share: $\text{GDP}_{\text{country}} / \text{GDP}_{\text{global}} \rightarrow \text{global energy share}$
- Energy Efficiency: $E_{\text{country}} / \text{GDP}_{\text{country}}$

- **Output (1960-2100)**

- How much of the world's future energy demand these 42 countries account for:
 $\sum_{42} E_{\text{country}} / E_{\text{global}} = E_{42} / E_{\text{global}}$

- **$R^2 = 0.997$**

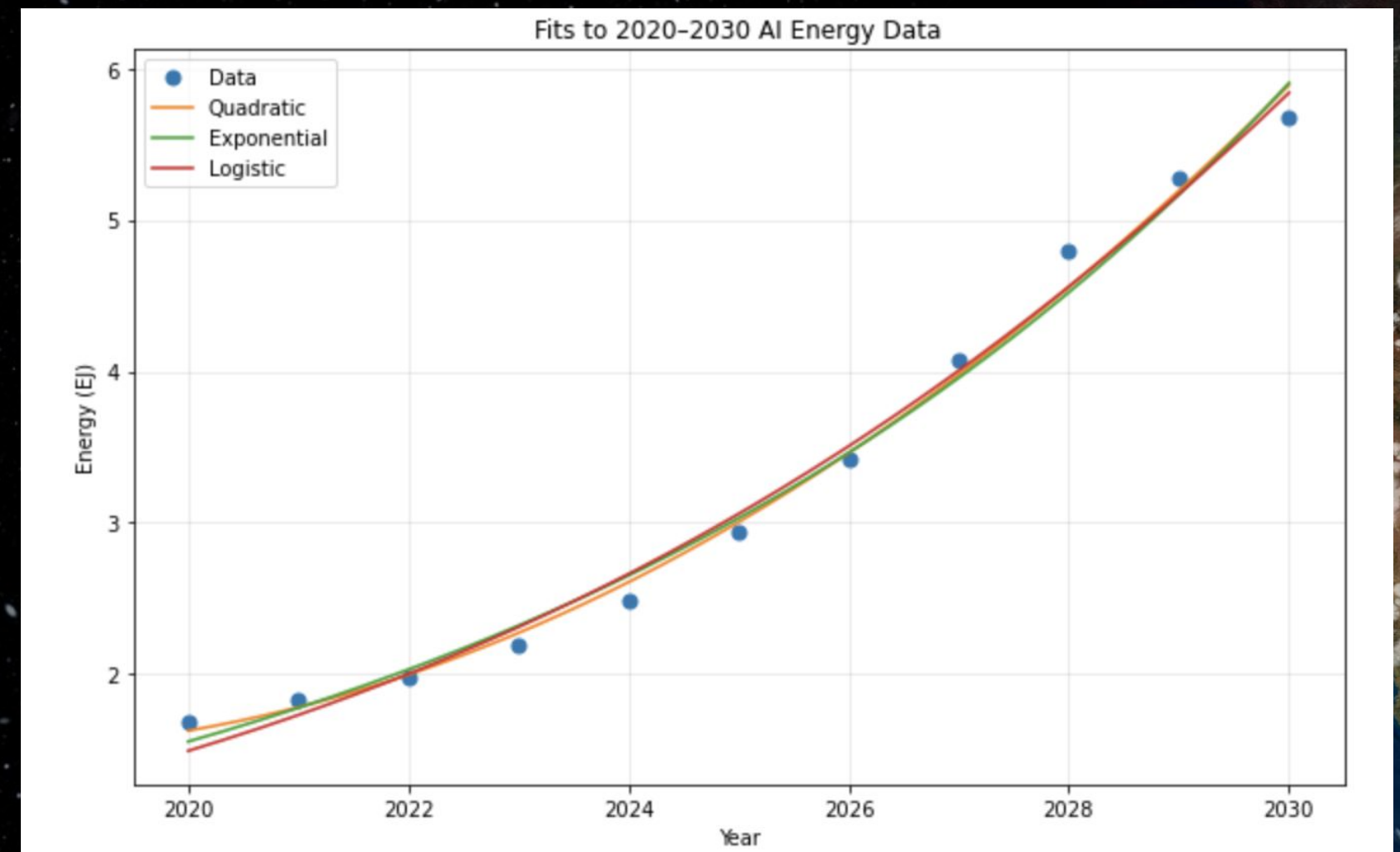


External factors

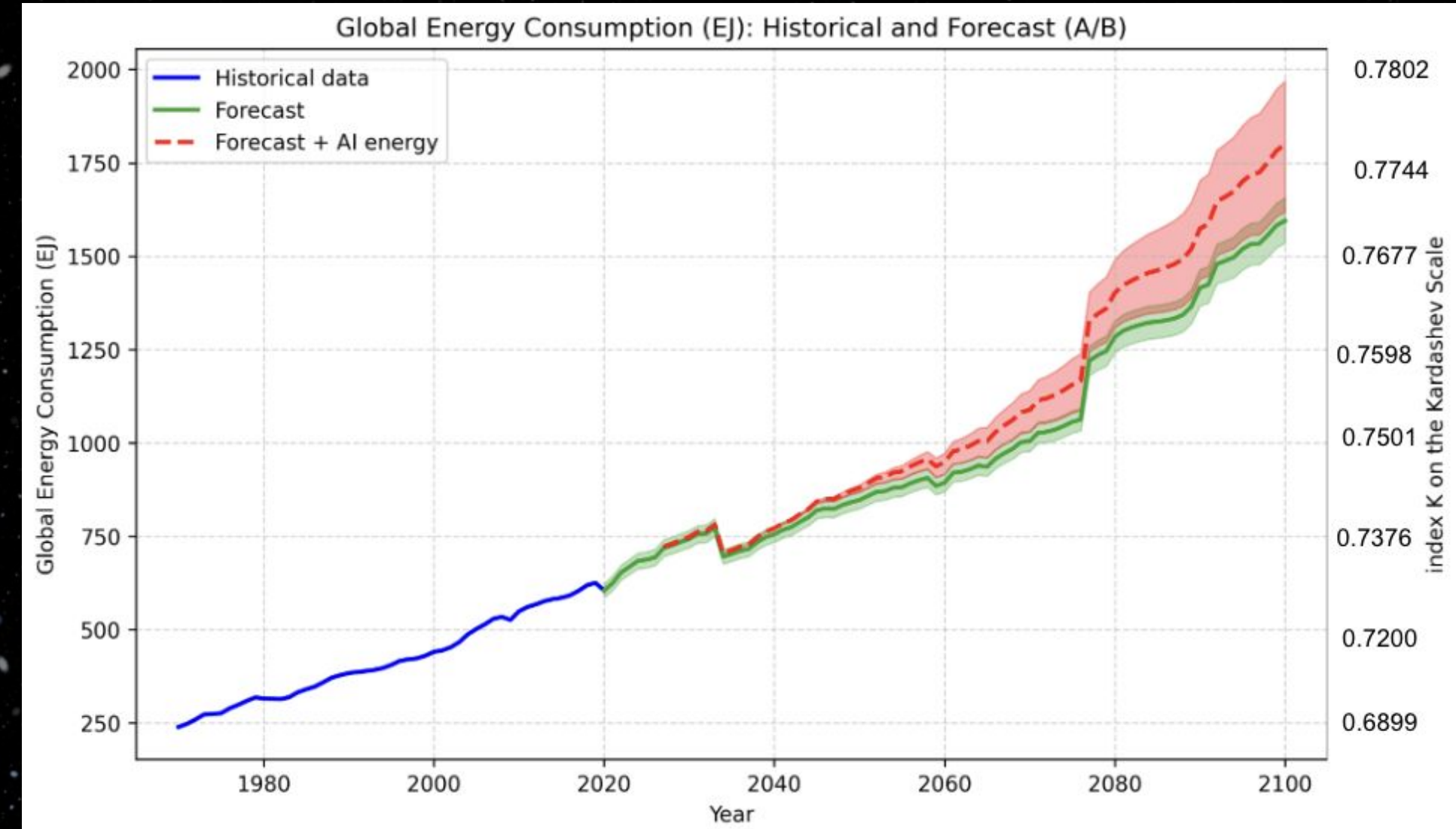
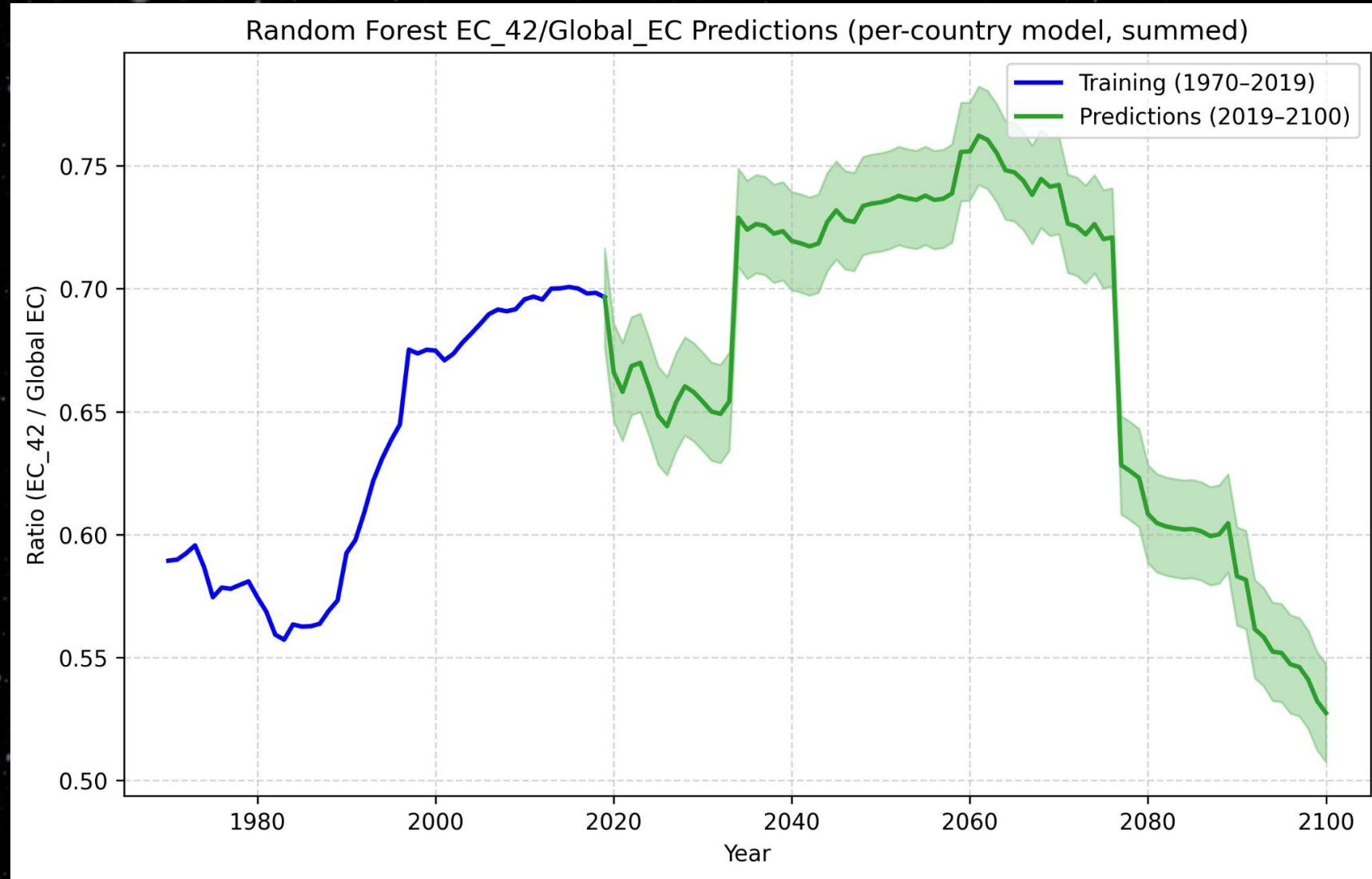
- Development of nuclear fusion , extreme climate mitigation policies

- **Impact of AI → demand-side force**

- International Energy Agency projections for global demand from AI data centers
 - Different fits of projected AI energy demand (least squares)
 - Quadratic best fit $R^2 = 0.99$, gompertz and logistic fits taken as "uncertainty"
- added to global energy prediction



Energy Ratio & Final Forecast Results



- 0.04 uncertainty assigned based on variation in historical data
- Finer features than ARIMA predictions
- $R^2 = 0.997$

K prediction for 2100

AI growth track : $\sim 0.776 \pm 0.01$

non-AI track : $\sim 0.770 \pm 0.002$

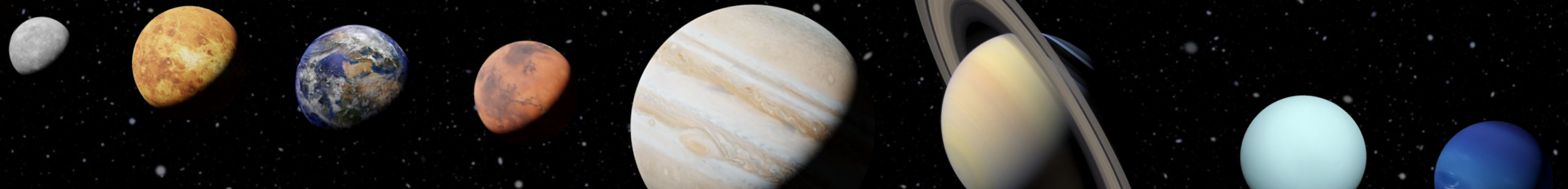


Astrobiology implications

- Will not achieve “planetary civilization” status by 2100
- Results are sensitive to many unpredictable factors such as technology, economics, geopolitics
- ML can be used to predict energy consumption of human civilization from existing data, but with high uncertainty

Future work

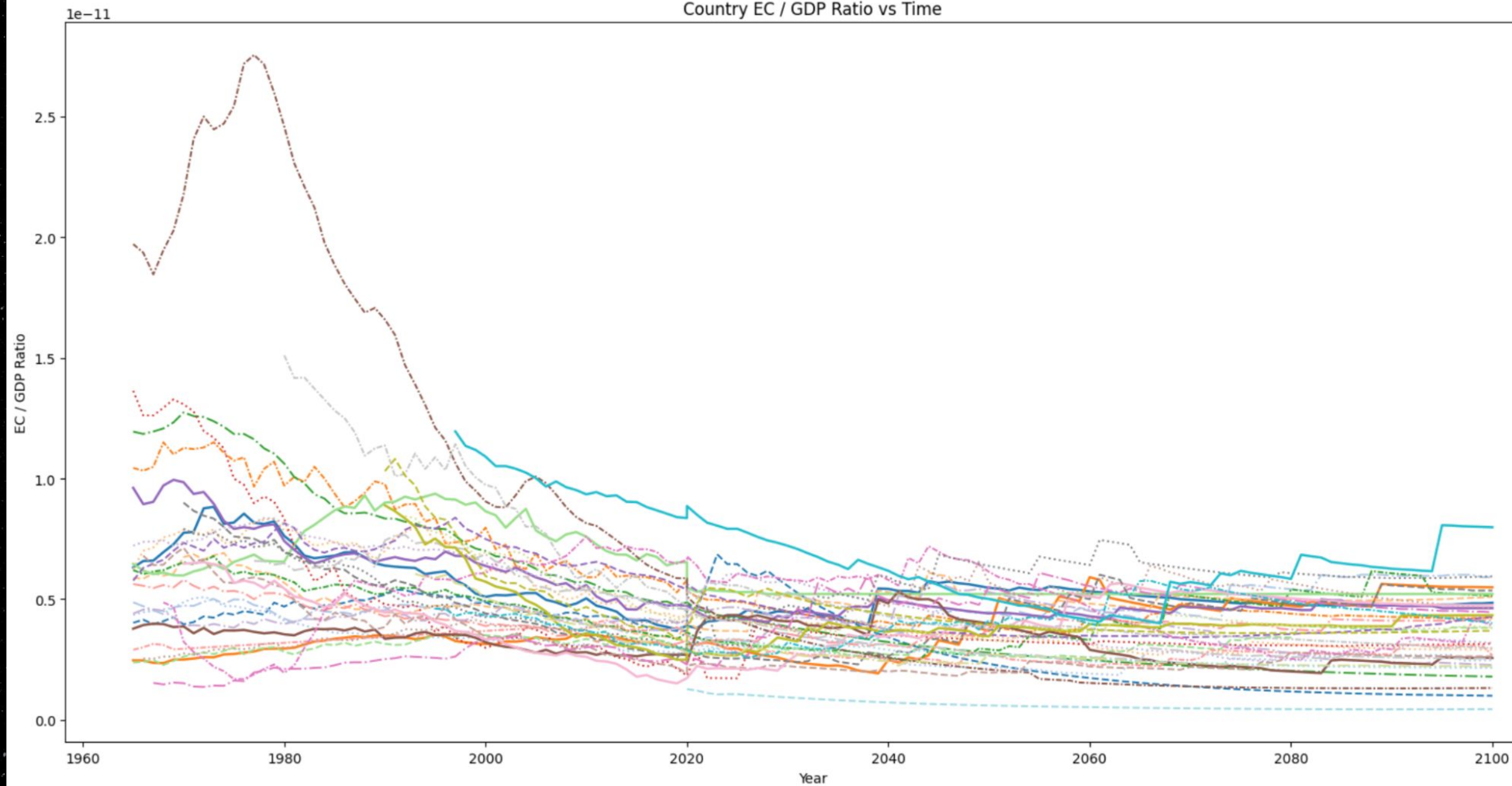
- Incorporating more data from different countries & time
- Add different “scenarios” with more robust modeling
- Extend further out into future





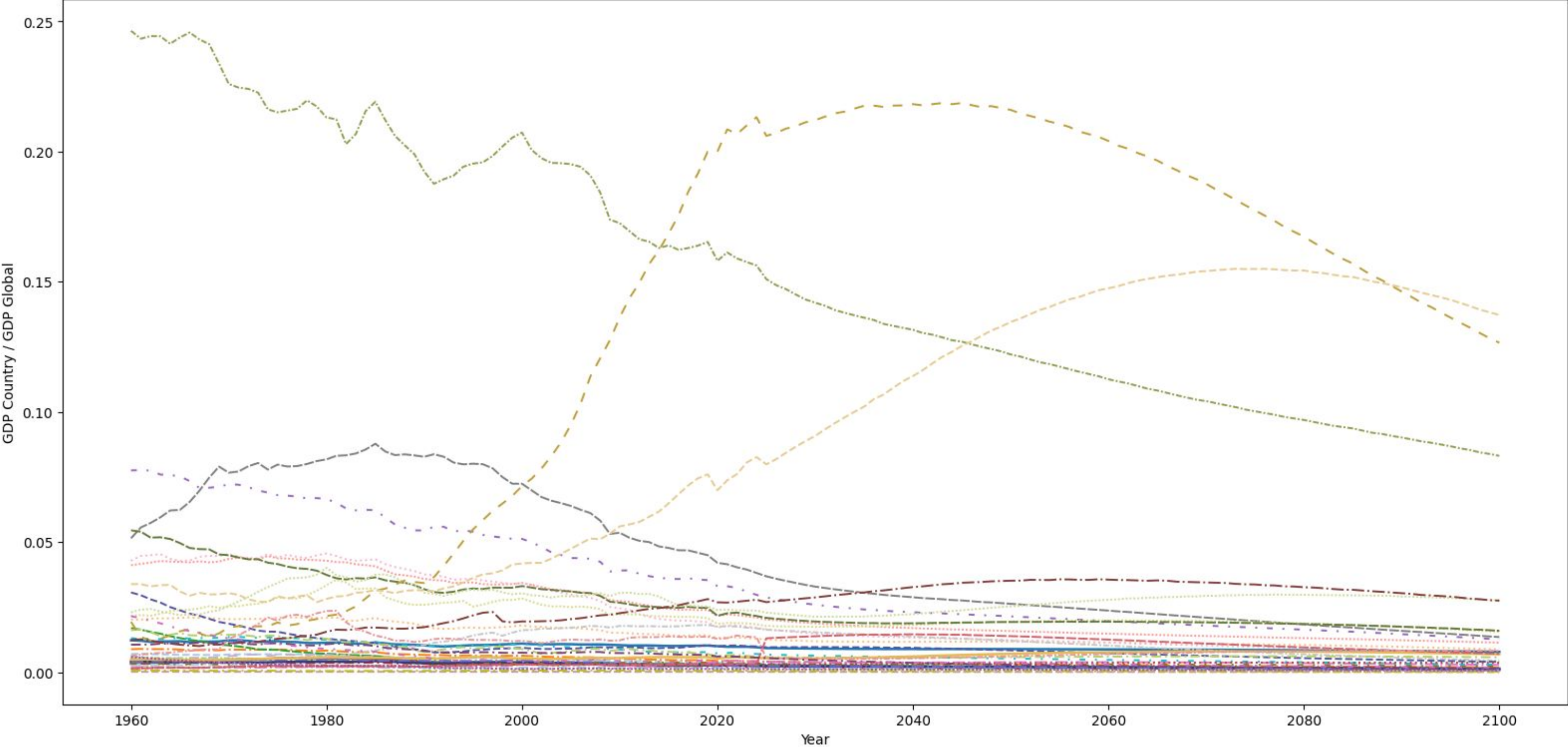
Thank you!

Country EC / GDP Ratio vs Time



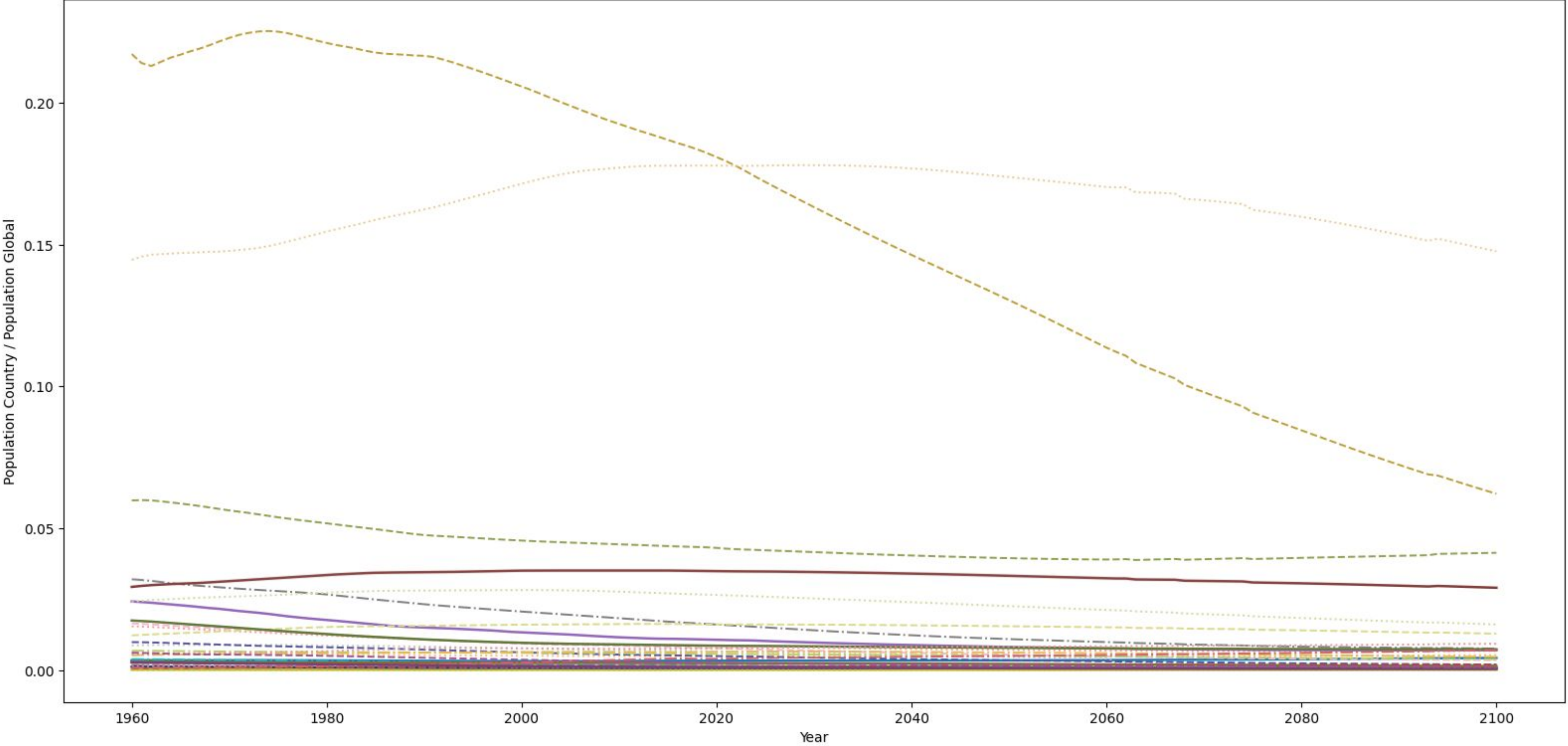
Netherlands	Japan	Luxembourg	Argentina	Indonesia	New Zealand	Hungary
India	Finland	Austria	China	Saudi Arabia	Bulgaria	Israel
Italy	United States of America	Greece	Colombia	Ireland	Romania	Canada
Chile	France	Belgium	Denmark	United Kingdom of Great Britain and Northern Ireland	Poland	Iceland
Norway	South Africa	Korea, Republic of	Brazil	Germany	Czechia	Slovakia
Mexico	Portugal	Australia				

Country Share of Global GDP vs Time



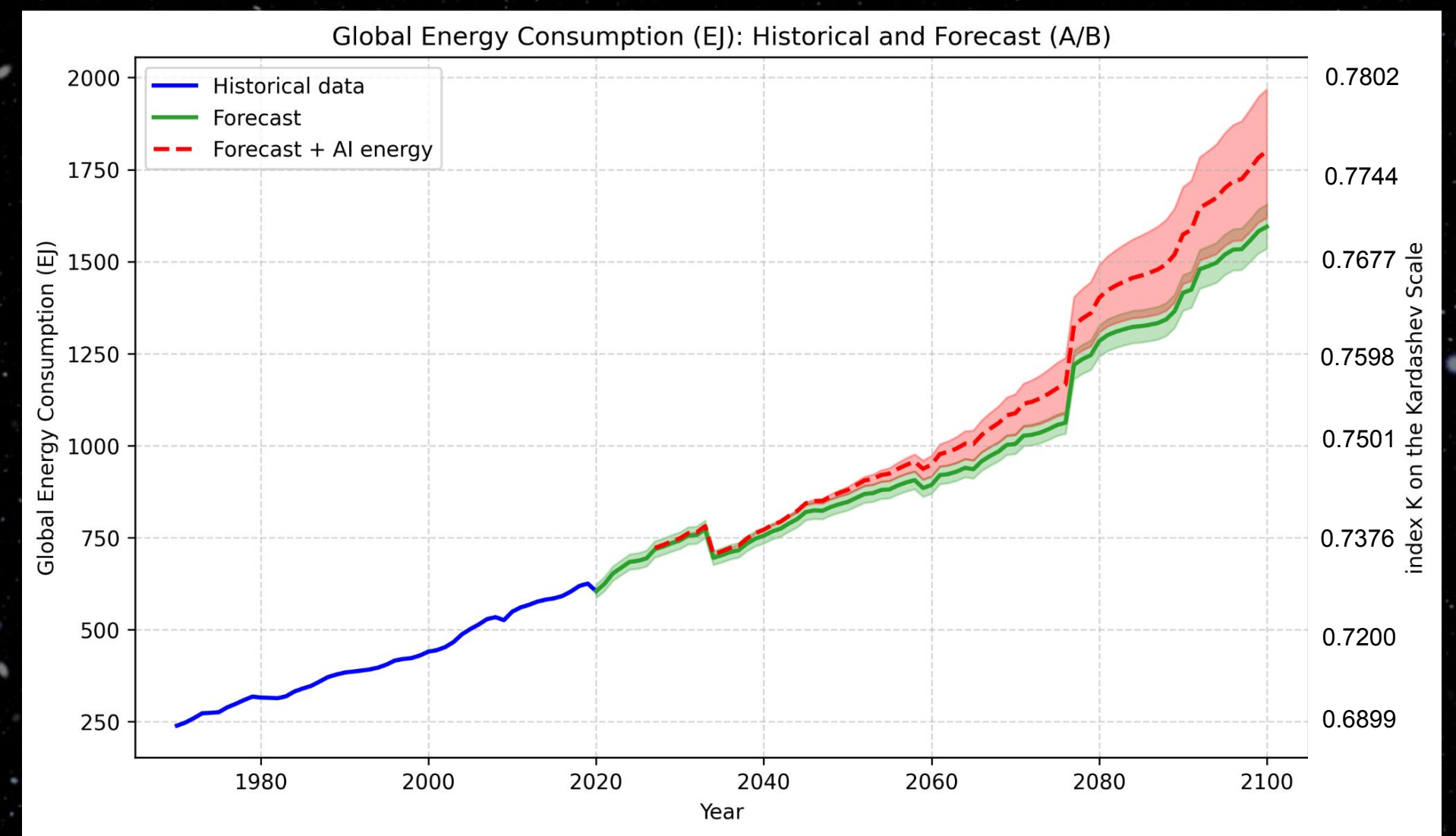
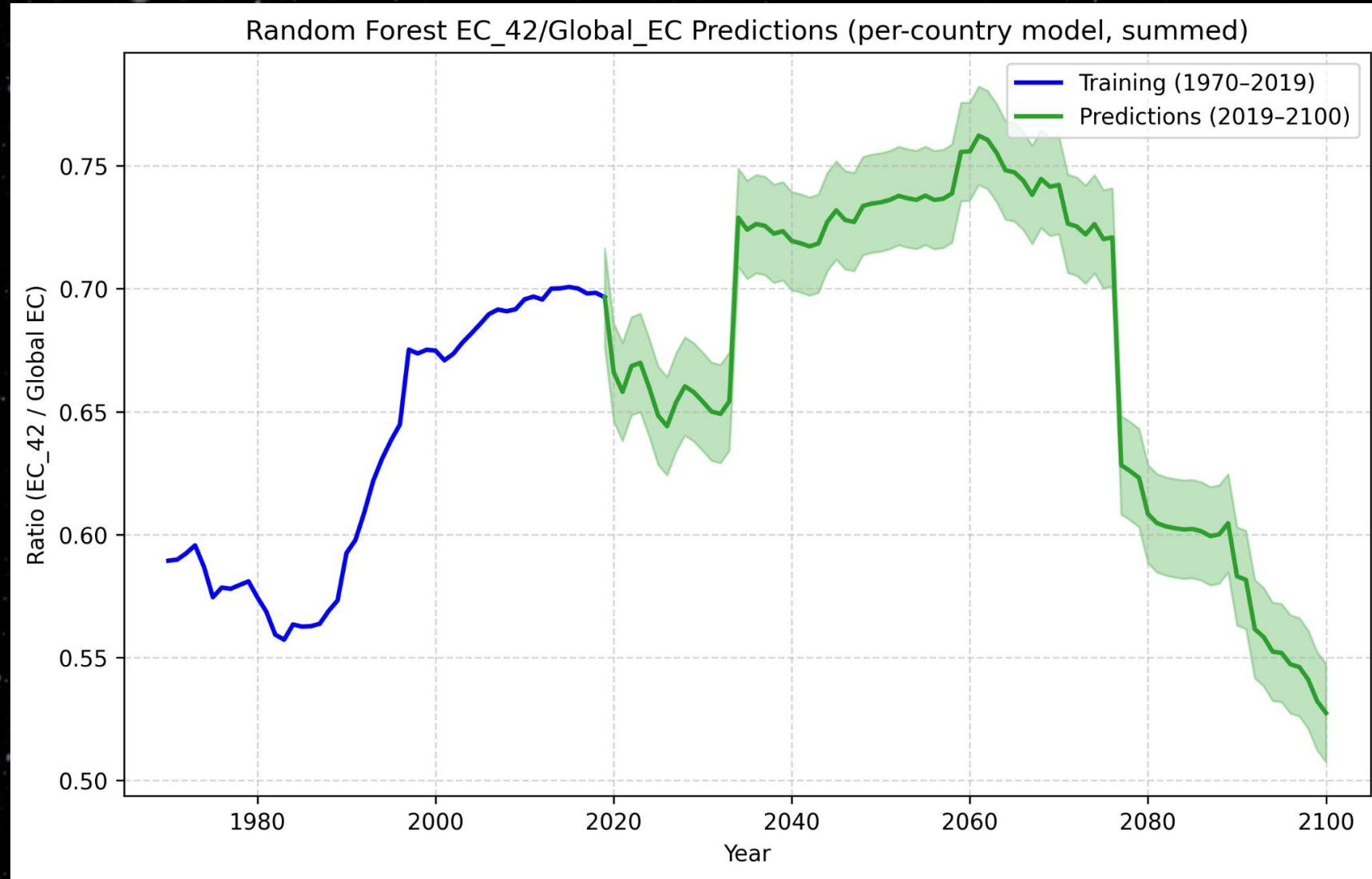
- | | | | | | | |
|-----------|---------|--------------------|-------------|--|-----------|--------------|
| Australia | Finland | Ireland | Netherlands | United Kingdom of Great Britain and Northern Ireland | China | Israel.1 |
| Austria | France | Italy | New Zealand | United States of America | Colombia | Saudi Arabia |
| Belgium | Germany | Japan | Norway | Argentina | India | South Africa |
| Canada | Greece | Korea, Republic of | Poland | Brazil | Indonesia | Bulgaria |
| Czechia | Hungary | Luxembourg | Portugal | Chile | Israel | Romania |
| Denmark | Iceland | Mexico | Slovakia | | | |

Population Ratio vs Time



- | | | | | | | |
|-----------|---------|------------|-------------|-----------|-----------|--------------|
| Australia | Finland | Ireland | Netherlands | Slovakia | Chile | Israel |
| Austria | France | Italy | New Zealand | UK | China | Saudi Arabia |
| Belgium | Germany | Japan | Norway | USA | Colombia | South Africa |
| Canada | Greece | Korea | Poland | Argentina | India | Bulgaria |
| Czechia | Hungary | Luxembourg | Portugal | Brazil | Indonesia | Romania |
| Denmark | Iceland | Mexico | | | | |

Energy Ratio & Final Forecast Results



- 0.04 uncertainty assigned manually due to variation in historical data
- Finer features than ARIMA predictions
- “discontinuities” may reflect large changes in input data which our model is sensitive to